

4. Apache Spark MLlib

Apache Spark MLlib is a machine learning library that has been designed to make practical machine learning scalable and easy. It comprises common learning algorithms and utilities, including classification, regression, clustering, collaborative filtering, dimensionality reduction as well as lower-level optimisation primitives and higher-level pipeline APIs.

Spark MLlib is regarded as a distributed machine learning framework on top of the Spark Core which, due to the distributed memory-based Spark architecture, is almost nine times as fast as the disk-based implementation used by Apache Mahout.

Given below are the various common machine learning and statistical algorithms that have been implemented and included with MLlib.

- **Classification:** Logistic regression, naïve Bayes
- **Regression:** Generalised linear regression, survival regression
- Decision trees, random forests, and gradient-boosted trees
- **Recommendation:** Alternating least squares (ALS)
- **Clustering:** K-means, Gaussian mixtures (GMMs)
- **Topic modelling:** Latent Dirichlet allocation (LDA)
- Frequency item sets, association rules and sequential pattern mining

Features

- **Easy to use:** MLlib easily fits into Spark's API and interoperates with NumPy in Python and R libraries.
- **Best performance:** Spark MLlib is excellent in performance – almost 100 times better than MapReduce. MLlib contains high-quality algorithms that leverage iteration and gives better results than one-pass approximations.
- **Multi-platform support:** MLlib can run on Hadoop, Apache Mesos, Kubernetes, a standalone system or in the cloud.

Official website: <https://spark.apache.org/mllib/>

Latest version: 2.1.3

5. Oryx 2

Oryx 2 is the result of Lambda architecture built on Apache Spark and Apache Kafka for real-time large-scale machine learning. It is designed for building applications and includes packaged, end-to-end applications for collaborative filtering, classification, regression and clustering.

Oryx 2 comprises the following three tiers:

- **General Lambda architecture tier:** Provides batch, speed and serving layers, which are not specific to machine learning.
- Specialisation on top provides machine learning abstraction to hyper parameter selection, etc.
- End-to-end implementation of the same standard machine learning algorithms as an application (ALS, random decision forests, k-means).
Oryx 2 consists of the following layers of Lambda architecture as well as connecting elements.
- **Batch layer:** Used for computing new results from historical data and previous results.
- **Speed layer:** Produces and publishes incremental model updates from a stream of new data.
- **Serving layer:** Receives models and updates, and implements a synchronous API, exposing query operations on results.
- **Data transport layer:** Moves data between layers and takes input from external sources.

Official website: <http://oryx.io/>

Latest version: 2.6.0

6. Vowpal Wabbit

Vowpal Wabbit is an open source, fast, out-of-core learning system

sponsored by Microsoft and Yahoo! Research. It is regarded as an efficient, scalable implementation of online machine learning with techniques like online, hashing, allreduce, reductions, learning2search, active and interactive learning.

The platform supports the following:

- Multiple supervised and semi supervised learning problems – classification, regression and active learning
- Multiple learning options – OLS regression, single layer neural net, LDA, contextual-bandit, etc
- Multiple loss functions
- Multiple optimisation algorithms – SGD, BFGS, conjugate gradient
- Regularisation

Features

- **Input format:** The input format is highly flexible for any learning algorithm.
- **Speed:** The learning algorithm is extremely fast and it can be applied to learning problems with a sparse terafeature, i.e., 1012 sparse features.
- Highly scalable.
- **Feature pairing:** Subsets of features can be internally paired so that the algorithm is linear in the cross-product of the subsets. This is useful for resolving ranking problems.

Official website: <http://hunch.net/~vw/>

Latest version: 8.5.0 

References

- [1] <https://mahout.apache.org/>
- [2] <https://systemml.apache.org/>
- [3] <https://www.h2o.ai>
- [4] <https://spark.apache.org/mllib/>
- [5] <http://oryx.io/>
- [6] <http://hunch.net/~vw/>

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